

SET 2: CUSTOMER FEEDBACK ANALYSIS

Name	Mohamed Nazeer S
Register No.	192424084
Course Code & Name	DSA0601 – DATA HANDLING AND VISUALIZATION

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

QUESTION 1 – HISTOGRAM: DISTRIBUTION OF CUSTOMER AGES

AIM

To create a histogram to represent the distribution of customer ages and understand the frequency of customers in different age ranges.

ALGORITHM

1. Import the Matplotlib library.
2. Store customer age values in a list.
3. Create a histogram using the hist() function.
4. Specify suitable bins for grouping customer ages.
5. Label the X-axis as Customer Age and Y-axis as Frequency.
6. Add a suitable title to the chart.
7. Display the histogram.

PROCEDURE

Open Google Colab or a Python environment. Enter the customer ages directly in the program. Execute the code. The histogram groups the ages into intervals and displays the frequency of customers in each age range.

CODE

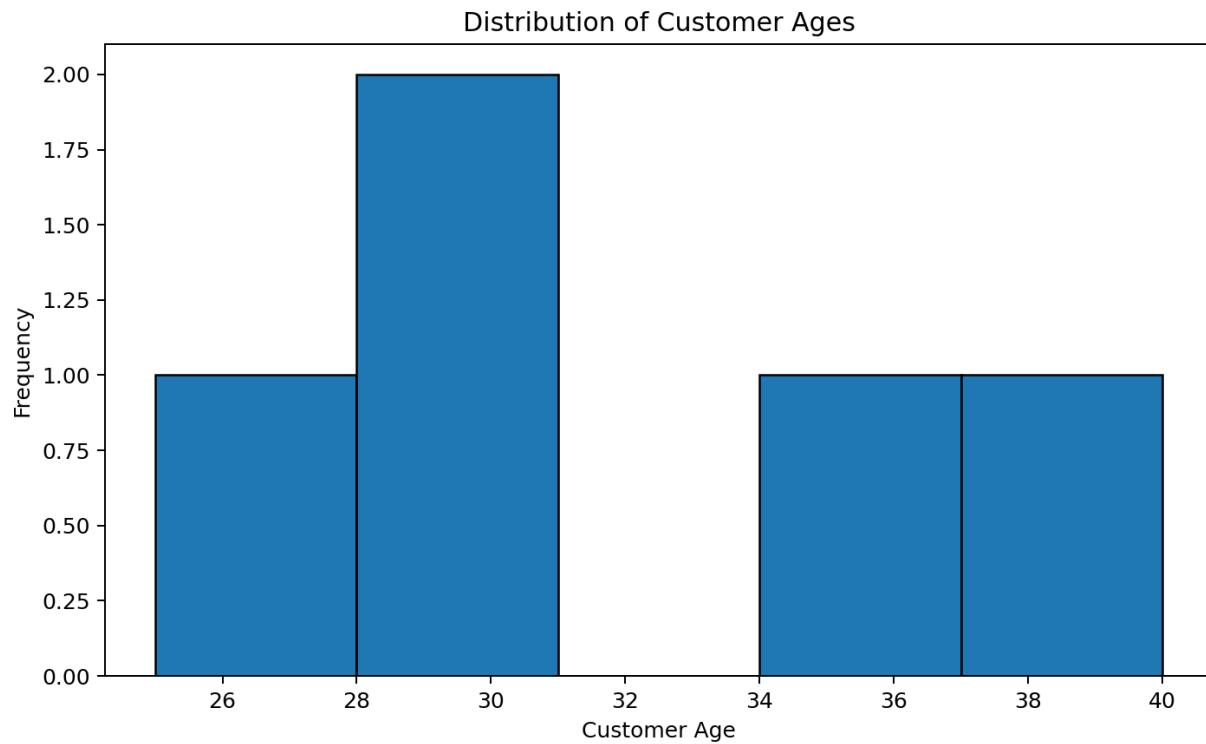
```
import matplotlib.pyplot as plt

ages = [25, 30, 35, 28, 40]

plt.figure(figsize=(8, 5))
plt.hist(ages, bins=5, edgecolor="black")
plt.xlabel("Customer Age")
```

```
plt.ylabel("Frequency")
plt.title("Distribution of Customer Ages")
plt.show()
```

OUTPUT SCREENSHOT



RESULT

Thus, the histogram representing the distribution of customer ages was created successfully. The chart shows how customers are distributed across the given age values from 25 to 40 years.

QUESTION 2 – PIE CHART: CUSTOMER SATISFACTION SCORES

AIM

To generate a pie chart to display the overall distribution of customer satisfaction scores and identify the proportion of each score.

ALGORITHM

1. Import Matplotlib and Pandas libraries.
2. Store the satisfaction scores in a list.
3. Count the frequency of each satisfaction score.
4. Create a pie chart using the pie() function.
5. Add score labels and percentage values.
6. Add a suitable title.
7. Display the pie chart.

PROCEDURE

Open Google Colab or a Python environment. Enter the satisfaction scores directly in the code. Run the program. The pie chart displays the percentage contribution of each satisfaction score to the total responses.

CODE

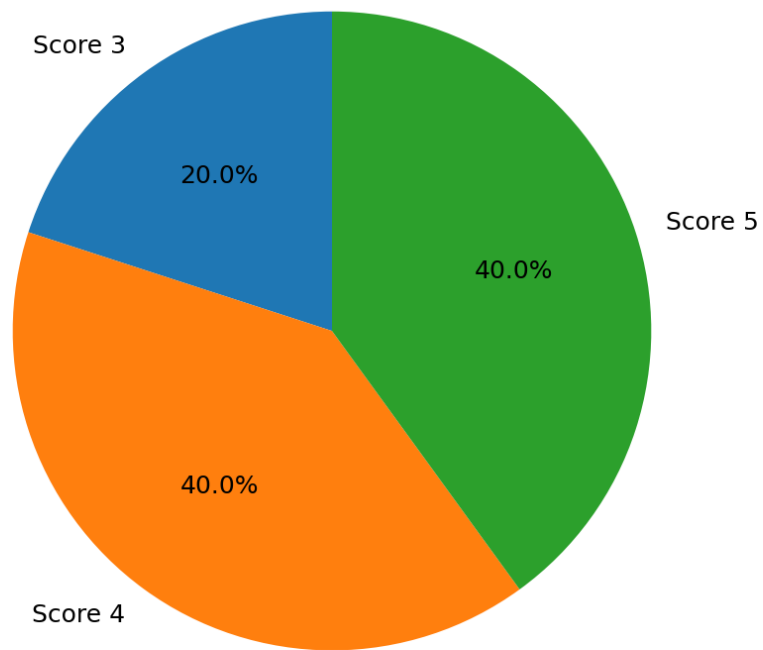
```
import matplotlib.pyplot as plt
import pandas as pd

scores = [4, 5, 3, 4, 5]
score_counts = pd.Series(scores).value_counts().sort_index()

plt.figure(figsize=(7, 5))
plt.pie(
    score_counts.values,
    labels=["Score " + str(x) for x in score_counts.index],
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Distribution of Customer Satisfaction Scores")
plt.show()
```

OUTPUT SCREENSHOT

Distribution of Customer Satisfaction Scores



RESULT

Thus, the pie chart for customer satisfaction scores was generated successfully. Scores 4 and 5 each represent 40% of customers, while score 3 represents 20%. Therefore, most customers show high satisfaction.

QUESTION 3 – STACKED BAR CHART: SATISFACTION BY AGE GROUP

AIM

To build a stacked bar chart to visualize the distribution of customer satisfaction scores across different customer age groups.

ALGORITHM

1. Import Pandas and Matplotlib libraries.
2. Create a DataFrame containing customer age and satisfaction score.
3. Divide customers into age groups 20–29, 30–39 and 40–49.
4. Create a frequency table using `crosstab()`.
5. Plot the frequency table as a stacked bar chart.
6. Label the axes and add a chart title and legend.
7. Display the stacked bar chart.

PROCEDURE

Open Google Colab or a Python environment. Enter the customer age and satisfaction score values directly in the code. Execute the program. Customers are grouped by age, and the stacked bars show the number of customers receiving each satisfaction score in every age group.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

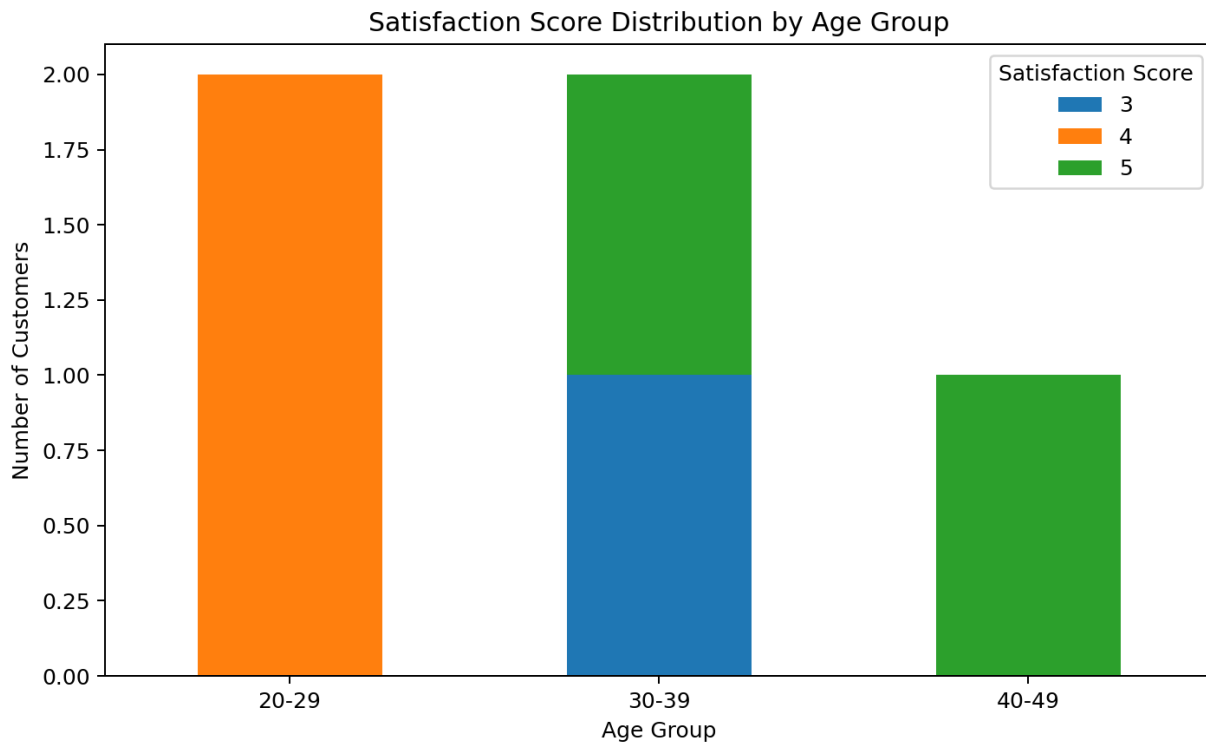
data = {
    "Age": [25, 30, 35, 28, 40],
    "Score": [4, 5, 3, 4, 5]
}
df = pd.DataFrame(data)

df["Age Group"] = pd.cut(
    df["Age"],
    bins=[20, 29, 39, 49],
    labels=["20-29", "30-39", "40-49"]
)

table = pd.crosstab(df["Age Group"], df["Score"])

table.plot(kind="bar", stacked=True, figsize=(8, 5))
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.title("Satisfaction Score Distribution by Age Group")
plt.legend(title="Satisfaction Score")
plt.xticks(rotation=0)
plt.show()
```

OUTPUT SCREENSHOT



RESULT

Thus, the stacked bar chart was created successfully. The 20–29 age group contains customers with satisfaction score 4, the 30–39 group contains scores 3 and 5, and the 40–49 group contains score 5. The chart helps compare satisfaction patterns among age groups.

QUESTION 4 – WORD CLOUD: OPEN-ENDED CUSTOMER FEEDBACK

AIM

To develop a word cloud from open-ended customer feedback and identify frequently occurring words that represent prevalent customer sentiments.

ALGORITHM

1. Import the WordCloud and Matplotlib libraries.
2. Store open-ended customer feedback as text.
3. Combine the feedback into a single text variable.
4. Create a WordCloud object.
5. Generate the word cloud from the feedback text.
6. Display the word cloud using imshow().
7. Remove the axes and add a suitable title.
8. Identify prominent words as common customer sentiments.

PROCEDURE

Open Google Colab or a Python environment. Install the wordcloud package if required using !pip install wordcloud. Enter the sample open-ended feedback directly in the code. Execute the program. Larger words in the output represent words that occur more frequently in customer feedback.

CODE

```
# Install once if required:
# !pip install wordcloud

from wordcloud import WordCloud
import matplotlib.pyplot as plt

feedback = """
excellent service very satisfied good experience friendly staff
quick support happy customer quality service excellent product
satisfied helpful staff good customer care easy process
great experience fast response reliable service happy satisfied
"""

wordcloud = WordCloud(
    width=900,
    height=500,
    background_color="white"
).generate(feedback)

plt.figure(figsize=(9, 5))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Customer Feedback Word Cloud")
plt.show()
```

OUTPUT SCREENSHOT

Customer Feedback Word Cloud



RESULT

Thus, the word cloud from open-ended customer feedback was developed successfully. Prominent words such as service, satisfied, excellent, good, happy and experience indicate that the sample customer feedback is predominantly positive.